

BRAKES

04
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04-00 OUTLINE

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BRAKE ABBREVIATIONS

dcf040000000t01

ABS	Antilock Brake System
CM	Control Module
HU	Hydraulic Unit
IDS	Integrated Diagnostic Software
LF	Left Front
LR	Left Rear
LSPV	Load Sensing Proportioning Valve
PID	Parameter Identification
PDS	Portable Diagnostic Software
RF	Right Front
RR	Right Rear
SW	Switch
4W-ABS	4-Wheel Antilock Brake System
4x2	4-wheel 2-drive
4x4	4-wheel 4-drive

04

BRAKE FEATURES

dcf040000000t02

Improved safety	4-wheel antilock brake system (4W-ABS) adopted - Rear antilock brake system (ABS) adopted
Improved braking force	- Tandem diaphragm power brake unit adopted - Load sensing proportioning valve (LSPV) has been adopted - Large diameter front disc brakes adopted - Two-piston type front disc brake caliper adopted (Hi-Rider, 4x4) - Wide width lining for the rear brake drum adopted - Vacuum pump adopted
Improved serviceability	Automatic adjustment mechanism rear brake (drum) has been adopted
Size and weight reduction	Integrated construction of the hydraulic unit (HU) and control module (CM) adopted for the ABS HU/CM (4W-ABS)

OUTLINE

BRAKE SPECIFICATIONS

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Item		Specification	
		4x2 (except Hi-Rider)	Hi-Rider, 4x4
Brake pedal	Type	Suspended design	
	Pedal lever ratio	3.68	
	Max. stroke (mm {in})	129 {5.08}	
Master cylinder	Type	Tandem (with level sensor)	
	Cylinder bore (mm {in})	25.4 {1.00}	
Front brake (disc)	Type	Ventilated disc	
	Cylinder bore (mm {in})	60.5 {2.38}	42.86 {1.687} x 2
	Pad dimensions (area x thickness) (mm ² x mm {in ² x in})	4,950 {7.673} x 10 {0.39}	5,500 {8.525} x 9 {0.35}
	Disc plate dimensions (outer diameter x thickness) (mm {in})	256 {10.1} x 24 {0.94}	289 {11.4} x 28 {1.1}
Rear brake (drum)	Type	Leading-trailing	
	Wheel cylinder bore (mm {in})	23.81 {0.9374}	
	Lining dimensions (width x length x thickness) (mm {in})	55 {2.2} x 271 {10.7} x 5.5 {0.22}	55 {2.2} x 296 {11.7} x 5.5 {0.22}
	Drum inner diameter (mm {in})	270 {10.6}	295 {11.6}
	Shoe clearance adjustment	Automatic adjuster	
Power brake unit	Type	Vacuum multiplier Tandem diaphragm	
	Outer diameter (mm {in})	188.4 {7.417} + 215.2 {8.472}	
Rear wheel braking force control device	Type	Load sensing proportioning valve	
Parking brake	Type	Mechanical two-rear-wheel control	
	Operation system	Stick lever type	
Brake fluid	Type	SAE J1703, FMVSS 116 DOT-3	

04-02B ON-BOARD DIAGNOSTIC [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ON-BOARD DIAGNOSTIC SYSTEM

OUTLINE [4W-ABS] 04-02B-1

ON-BOARD DIAGNOSTIC SYSTEM

FUNCTION [4W-ABS] 04-02B-2

ON-BOARD DIAGNOSTIC SYSTEM

PID/DATA MONITOR FUNCTION
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ON-BOARD DIAGNOSTIC SYSTEM

ACTIVE COMMAND MODES

FUNCTION [4W-ABS] 04-02B-3

ON-BOARD DIAGNOSTIC SYSTEM

EXTERNAL TESTER COMMUNICATION

FUNCTION [4W-ABS] 04-02B-4

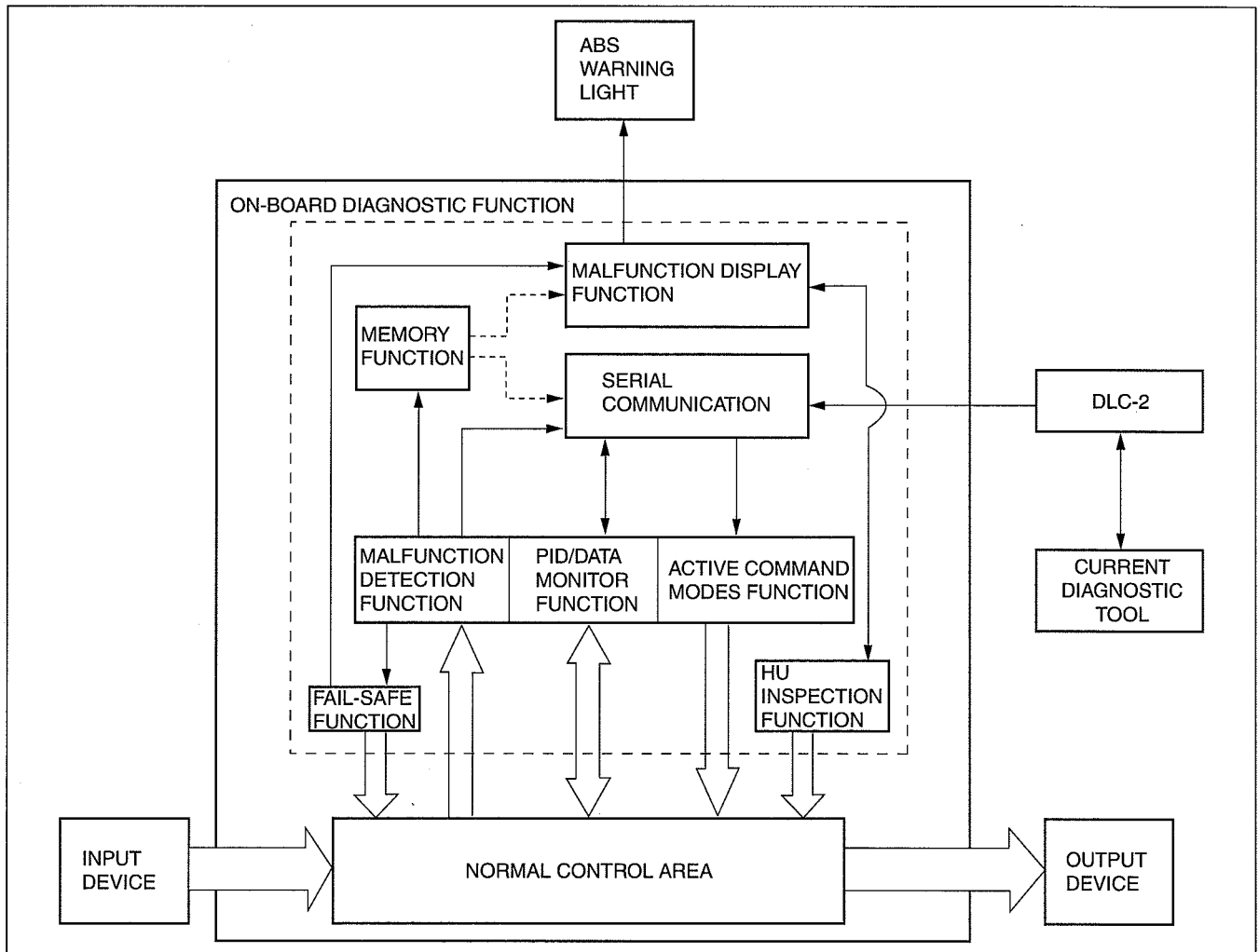
DLC-2 CONSTRUCTION [4W-ABS] 04-02B-5

ON-BOARD DIAGNOSTIC SYSTEM OUTLINE [4W-ABS]

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- The on-board diagnostic system consists of a malfunction detection system that detects abnormalities in input/output signals when the ignition switch is at the ON position, a data monitor function that reads out specified input/output signals and a simulation function that allows for override operation of output parts (such as solenoid valves).
- The data link connector 2 (DLC-2), which groups together all the connectors used for malfunction diagnosis and detecting/repair into a single location, has been adopted, thereby improving serviceability. Diagnosis is performed by connecting the current diagnostic tool to the DLC-2.
- In addition to DTC read-out, the current diagnostic tool is used to clear DTCs using the display screen of the diagnostic tester, and to access the PID/data monitor and simulation functions, providing enhanced malfunction diagnosis and improved serviceability.

Block Diagram



DCF402BTB001

ON-BOARD DIAGNOSTIC [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ON-BOARD DIAGNOSTIC SYSTEM FUNCTION [4W-ABS]

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Malfunction Detection Function

- The malfunction detection function detects malfunctions in the input/output signal system of the ABS HU/CM when the engine switch is at the ON position.
- When the ABS HU/CM are started up, the following malfunction detections are performed.
 - The ABS warning light illuminate for **approx. 3 s** when the engine switch is turned to the ON position. At the same time, the fail-safe relay is operated, and the input/output signals of each part is monitored for malfunction diagnosis. After starting to drive, the first time the vehicle speed is **approx. 12 km/h {7.5 mph} or more** the pump motor is operated and malfunction diagnosis is performed again.
- When malfunctions are detected, the corresponding lights are illuminated to alert the driver. Using the external tester communication function, DTCs can be output through the KLN of the DLC-2. At the same time, malfunction detection results are sent to the memory and fail-safe functions.

Memory Function

- The memory function stores DTCs of malfunctions in input/output signal systems. With this function, once a DTC is stored it is not cleared after the engine switch has been turned off (LOCK position), even if the malfunctioning signal system has returned to normal.
- Since the ABS HU/CM has a built-in non-volatile memory, DTCs are not cleared even if the battery is removed. Therefore, it is necessary to clear the memory after performing repairs. Refer to the Workshop Manual for the DTC clearing procedure.

Fail-safe Function

- When the malfunction detection function determines a malfunction, each light illuminates to advise the driver. At this time, the fail-safe function controls the ABS as shown in the fail-safe function table.

Fail-safe Function Malfunction Contents

Malfunction location	DTC number	Fail-safe function	
		Warning light illumination status	Control status
		ABS warning light	ABS control
Power supply system	B1317	Illuminated*1	Control disabled*2
	B1318		
ABS HU/CM system	B1342	Illuminated	Control disabled
Brake switch signal system	B1484	Not illuminated	Control enabled
Pump motor, motor relay systems	C1095	Illuminated	Control disabled
	C1096		
ABS wheel-speed sensor system	C1145	Illuminated	Control disabled
	C1155		
	C1165		
	C1175		
Fail-safe relay system	C1186	Illuminated	Control disabled
Solenoid valve system	C1194	Illuminated	Control disabled
	C1198		
	C1202		
	C1206		
	C1210		
	C1214		
ABS wheel-speed sensor (slip monitor) system	C1222	Illuminated	Control disabled
ABS wheel-speed sensor/ABS sensor rotor systems	C1233	Illuminated	Control disabled
	C1234		
	C1235		
	C1236		
Incorrect ABS HU/CM installed	C1414	Illuminated	Control disabled
G sensor system*3	C1730	Illuminated	Control disabled
	C1949		
	C1950		

*1 : If the ignition voltage returns to normal, the light goes out.

*2 : If the ignition voltage returns to normal, control is enabled.

*3 : Hi-Rider, 4x4

ON-BOARD DIAGNOSTIC [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ON-BOARD DIAGNOSTIC SYSTEM PID/DATA MONITOR FUNCTION [4W-ABS]

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- The PID/data monitor function is used for optionally selecting input/output signal monitor items preset in the ABS HU/CM and reading them out in real-time.

PID/data monitor item	Input/output part	Unit/Condition (Tester display)
ABS_LAMP	ABS warning light	On/Off
ABS_VOLT	Battery	V
ACCLMTR*	G sensor	G
BOO_ABS	Brake switch	On/Off
CCNTABS	Number of continuous DTCs	—
PMP_MOTOR	Pump motor	On/Off
RLY_PMP	Pump motor relay	On/Off
RLY_VLV	Solenoid valve relay	On/Off
V_LF_INL	LF inlet solenoid valve	On/Off
V_LF_OTL	LF outlet solenoid valve	On/Off
V_RF_INL	RF inlet solenoid valve	On/Off
V_RF_OTL	RF outlet solenoid valve	On/Off
V_Rear_INL	Rear inlet solenoid valve	On/Off
V_Rear_OTL	Rear outlet solenoid valve	On/Off
WSPD_LF	ABS wheel-speed sensor (LF)	KPH, MPH
WSPD_LR	ABS wheel-speed sensor (LR)	KPH, MPH
WSPD_RF	ABS wheel-speed sensor (RF)	KPH, MPH
WSPD_RR	ABS wheel-speed sensor (RR)	KPH, MPH

* : Hi-Rider, 4x4

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ON-BOARD DIAGNOSTIC SYSTEM ACTIVE COMMAND MODES FUNCTION [4W-ABS]

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- The active command modes function is used for optionally selecting active command modes items of input/output parts preset in the ABS HU/CM, and to operate them regardless of CM control.
- To protect the hydraulic unit interior, operate output related parts for only **10 s or less** when using the active command modes function.

Active Command Modes Table

Command name	Output part name	Operation	Operation condition
PMP_MOTOR	Pump motor	On/Off	Engine switch at ON
V_LF_INL	LF inlet solenoid valve		
V_LF_OTL	LF outlet solenoid valve		
V_Rear_INL	Rear inlet solenoid valve		
V_Rear_OTL	Rear outlet solenoid valve		
V_RF_INL	RF inlet solenoid valve		
V_RF_OTL	RF outlet solenoid valve		

ON-BOARD DIAGNOSTIC [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ON-BOARD DIAGNOSTIC SYSTEM EXTERNAL TESTER COMMUNICATION FUNCTION [4W-ABS]

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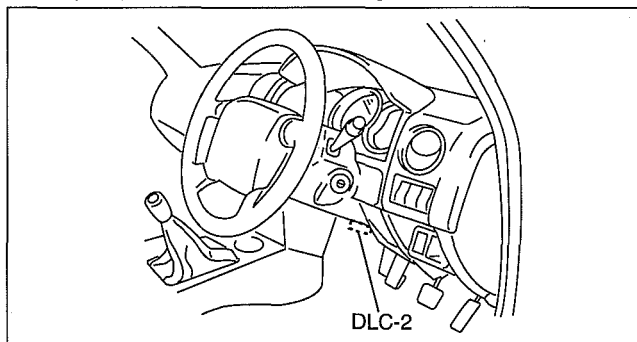
- The external tester communication function enables communication of diagnostic data (DTC read-outs, input/output signal read-outs, and operation of input/output parts) between the ABS HU/CM and an external tester.

Connections and Communication Contents

	External tester	
	Current diagnostic tool	
	Connection	Communication method
On-board diagnostic (malfunction detection) function	Input/output: KLN	Serial communication
PID/DATA monitor function	Input/output: KLN	Serial communication
Active command modes function	Input/output: KLN	Serial communication

Serial communication

- Serial communication (two-way communication) allows for multiple data to be sent and received instantly along the same line.
- By connecting the current diagnostic tool to the DLC-2, diagnostic data can be sent and received between the current diagnostic tool and the ABS HU/CM using the KLN terminals (within the DLC-2).
- The ABS HU/CM receives the command signals of the malfunction detection function, PID/data monitor function, and the active command modes function from the current diagnostic tool, and sends DTCs and data regarding the operating condition and status of each input/output part to the current diagnostic tool.



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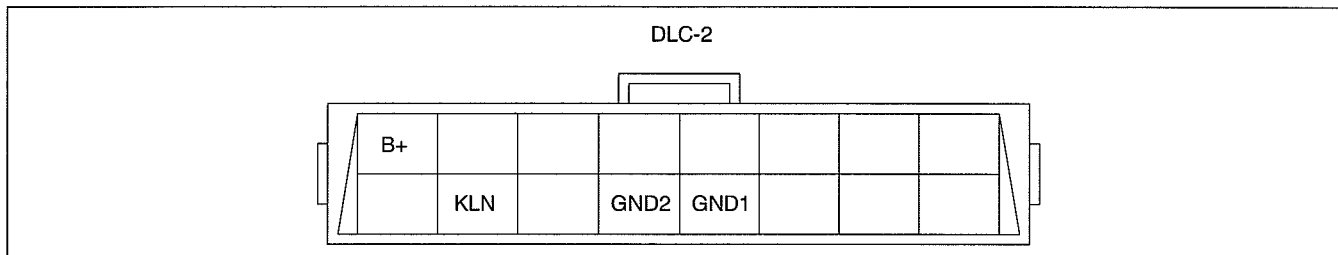
Diagnostic function name	Signal received	Signal sent
Malfunction detection function	DTC verification signal	DTC
PID/data monitor function	Command signal to read selected monitor item	Monitored data for requested monitor item
Active command modes function	Operation command signal for selected active command modes item	Input/output part name

ON-BOARD DIAGNOSTIC [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

DLC-2 CONSTRUCTION [4W-ABS]

dcf040243750t09

- A connector (DLC-2) conforming to International Organization for Standardization (ISO) standards has been added.
- Shape and terminal arrangement as stipulated by the ISO 15031-3 (SAE J1962) international standard has been adopted for this connector. The connector has a 16-pin construction that includes the KLN, GND1, GND2 and B+ terminals.



DBR402ZTB002

Terminal	Function
KLN	Serial communication KLN terminal
GND1	Body ground terminal
GND2	Serial communication ground terminal
B+	Battery power supply terminal

(2)

(2)

(2)

04-11 CONVENTIONAL BRAKE SYSTEM

CONVENTIONAL BRAKE

SYSTEM OUTLINE 04-11-1

CONVENTIONAL BRAKE

SYSTEM STRUCTURAL VIEW..... 04-11-1

MASTER CYLINDER

CONSTRUCTION 04-11-2

POWER BRAKE UNIT

CONSTRUCTION 04-11-2

VACUUM PUMP CONSTRUCTION 04-11-2

LOAD SENSING PROPORTIONING

VALVE (LSPV) CONSTRUCTION04-11-3

LOAD SENSING PROPORTIONING

VALVE (LSPV) OPERATION04-11-4

FRONT BRAKE (DISC)

CONSTRUCTION.....04-11-6

REAR BRAKE (DRUM)

CONSTRUCTION04-11-7

REAR BRAKE (DRUM) OPERATION ...04-11-8

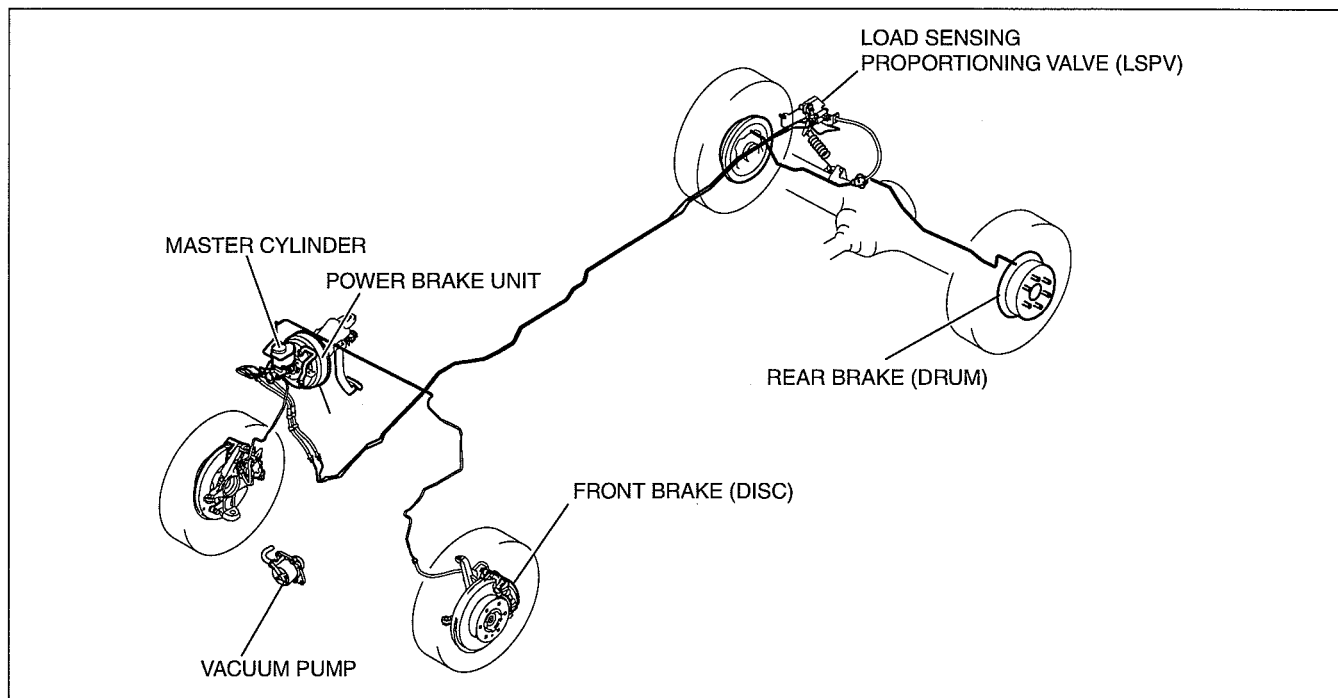
CONVENTIONAL BRAKE SYSTEM OUTLINE

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- A tandem-type master cylinder has been adopted, improving braking force.
- A large diameter, tandem diaphragm power brake unit has been adopted, improving braking force.
- A large diameter, ventilated disc-type front brake has been adopted, improving braking force.
- A two-piston type front disc brake caliper has been adopted, improving braking force. (Hi-Rider, 4x4)
- A wide width lining for the rear brake drum has been adopted, improving braking force.
- An Automatic adjustment mechanism rear brake (drum) has been adopted, improving serviceability
- A vacuum pump has been adopted, improving braking force.

CONVENTIONAL BRAKE SYSTEM STRUCTURAL VIEW

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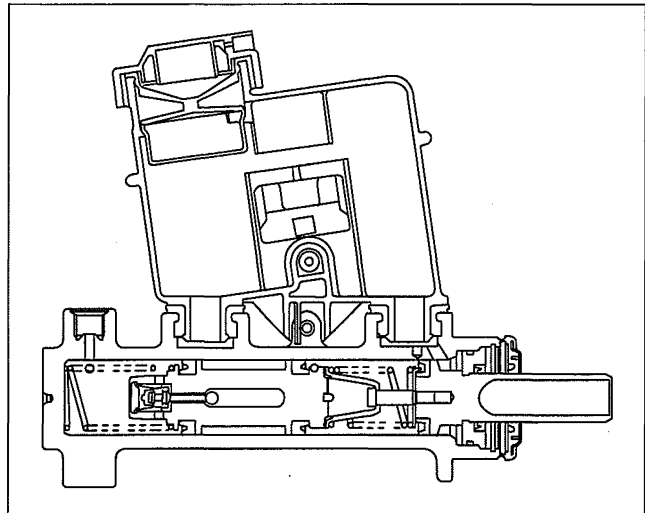
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CONVENTIONAL BRAKE SYSTEM

MASTER CYLINDER CONSTRUCTION

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- A tandem-type master cylinder with a **25.4 mm {1.00 in}** bore has been adopted, improving braking force.

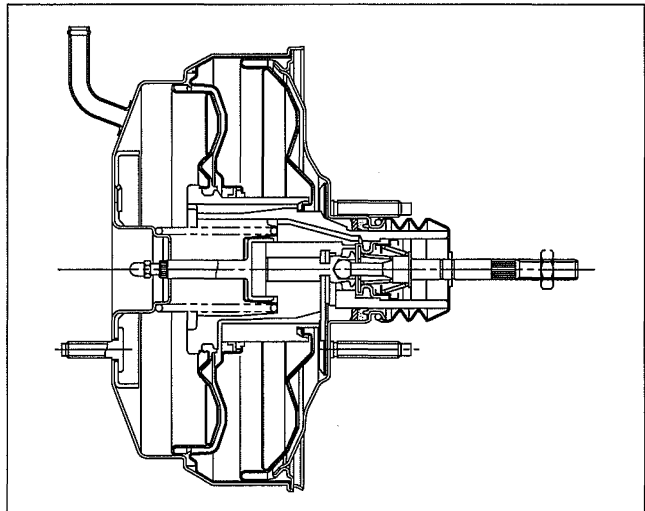


DBR411ZTB002

POWER BRAKE UNIT CONSTRUCTION

dcf041143800t01

- A tandem diaphragm type power brake unit has been adopted for all models, achieving compatibility between high braking performance and excellent brake feeling.

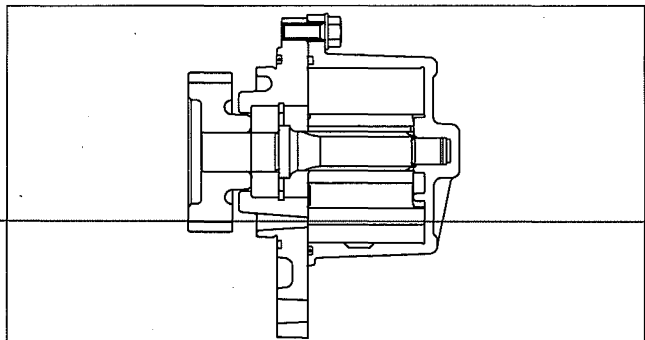


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VACUUM PUMP CONSTRUCTION

dcf041118777t01

- A gear-driven vacuum pump has been adopted, improving braking force.



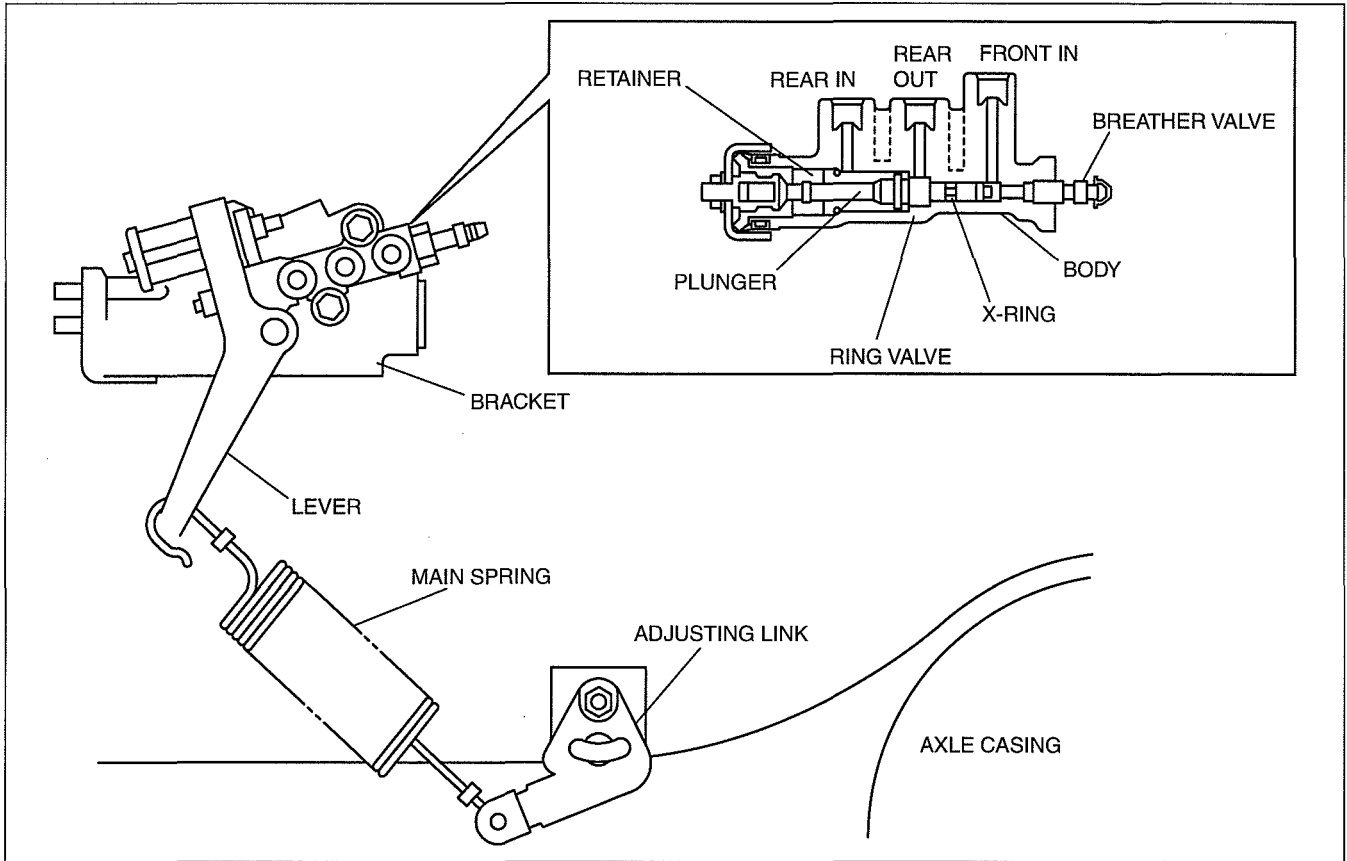
DBR411ZTB014

CONVENTIONAL BRAKE SYSTEM

LOAD SENSING PROPORTIONING VALVE (LSPV) CONSTRUCTION

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- This valve is to proportion the braking force to the front and rear wheels as ideally as possible for higher safety at the time of the brakes being applied, it controls the brake fluid pressure to the rear brakes according to the load on the rear axle so that the rear end of the vehicle may not spin due to locking of the rear wheels in advance of the fronts.
- This valve also incorporates a bypass passage that prevents pressure reduction so that rear wheels will still have plenty of braking force to stop the vehicle in the event of a front master cylinder malfunction.



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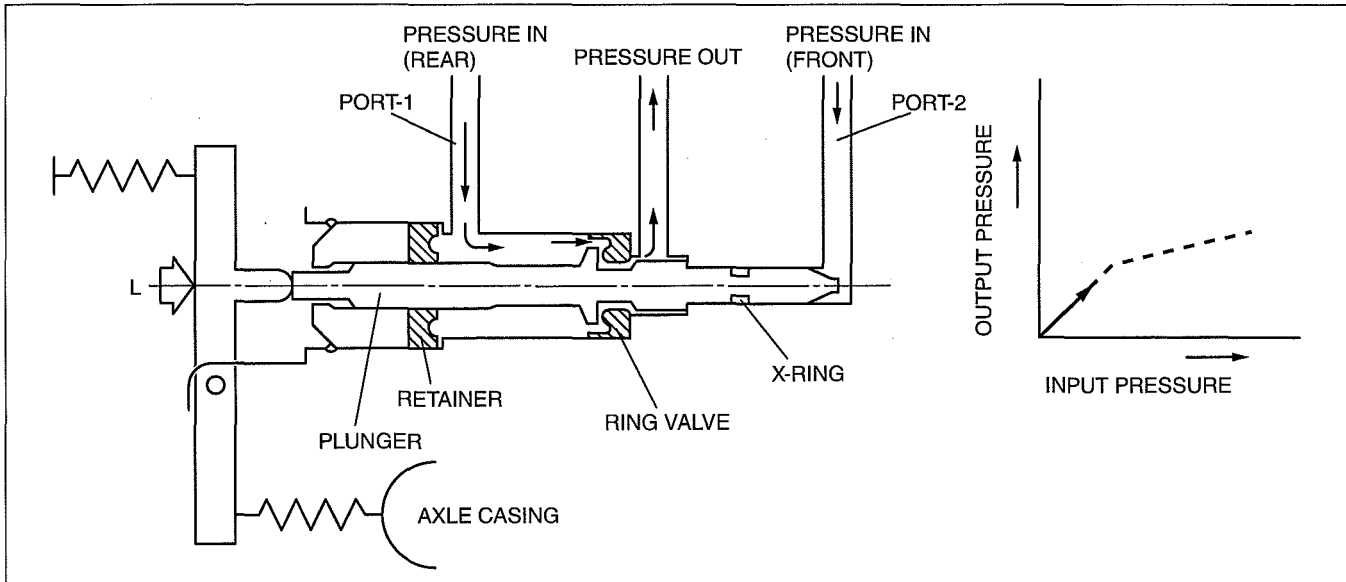
CONVENTIONAL BRAKE SYSTEM

LOAD SENSING PROPORTIONING VALVE (LSPV) OPERATION

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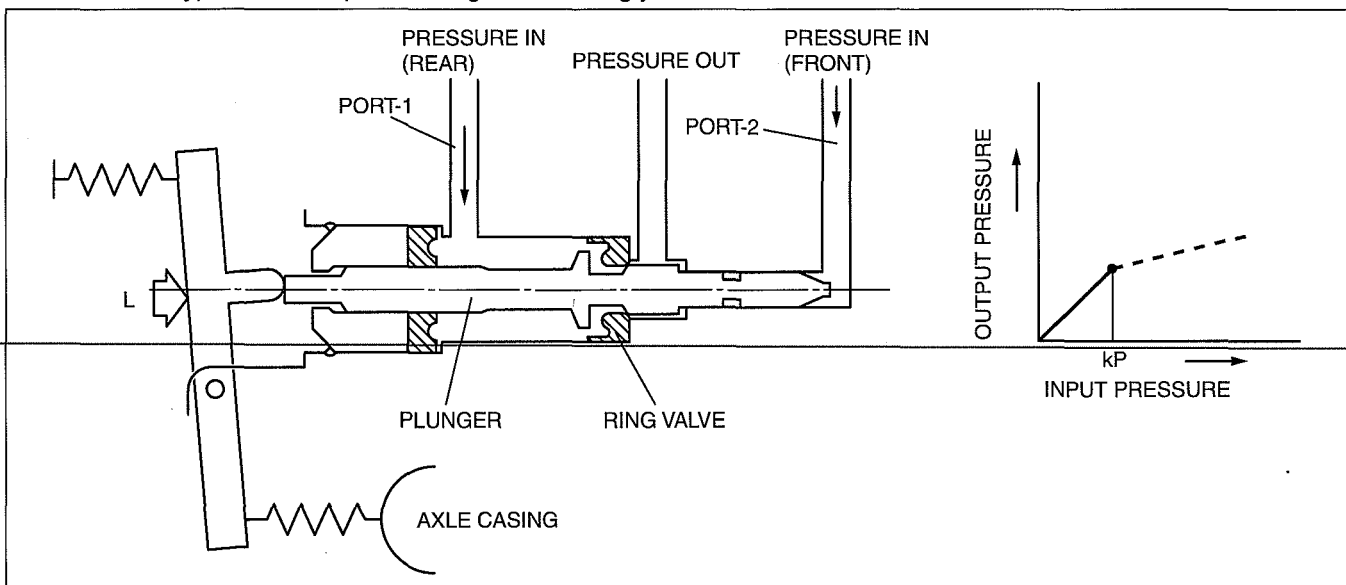
Braking When Vehicle Empty

- The hydraulic pressure generated at the rear section of the master cylinder is applied, via input port 1, to the hydraulic-pressure control section of the LSPV. The hydraulic pressure generated at the front section of the master cylinder is applied to the front brakes, and to the bypass section of the LSPV via input port 2. The rear wheel and front wheel hydraulic pressure are separated by the X-ring of the plunger. Since port 1 hydraulic pressure is normally equal to port 2 hydraulic pressure, the plunger is pressed against the cylinder base by L. When port 1 hydraulic pressure is applied to the plunger, the plunger tries to move to the left. But, because there is also at the same time force applied in the right direction by the lever link (force L), the plunger is held against the cylinder base. The result is that the hydraulic pressure at the output side is the same as at the input side when force L is higher than the input pressure.



DBR411ZTB009

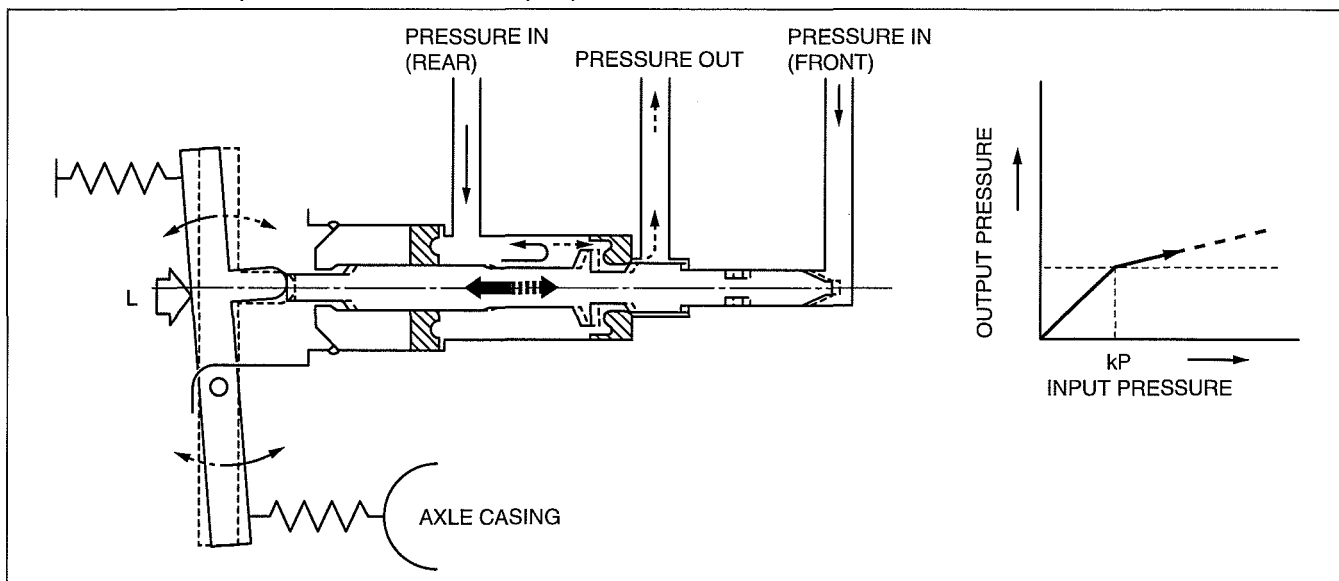
- When the input hydraulic pressure increases further and overcomes force L, the plunger is moved slightly leftward, sealing it against the ring valve. This temporarily separates the input side from the output side. The output pressure at this time is called the break point. If force L from the lever link changes (i.e. lowering of the vehicle body), this break point changes accordingly.



DBR411ZTB010

CONVENTIONAL BRAKE SYSTEM

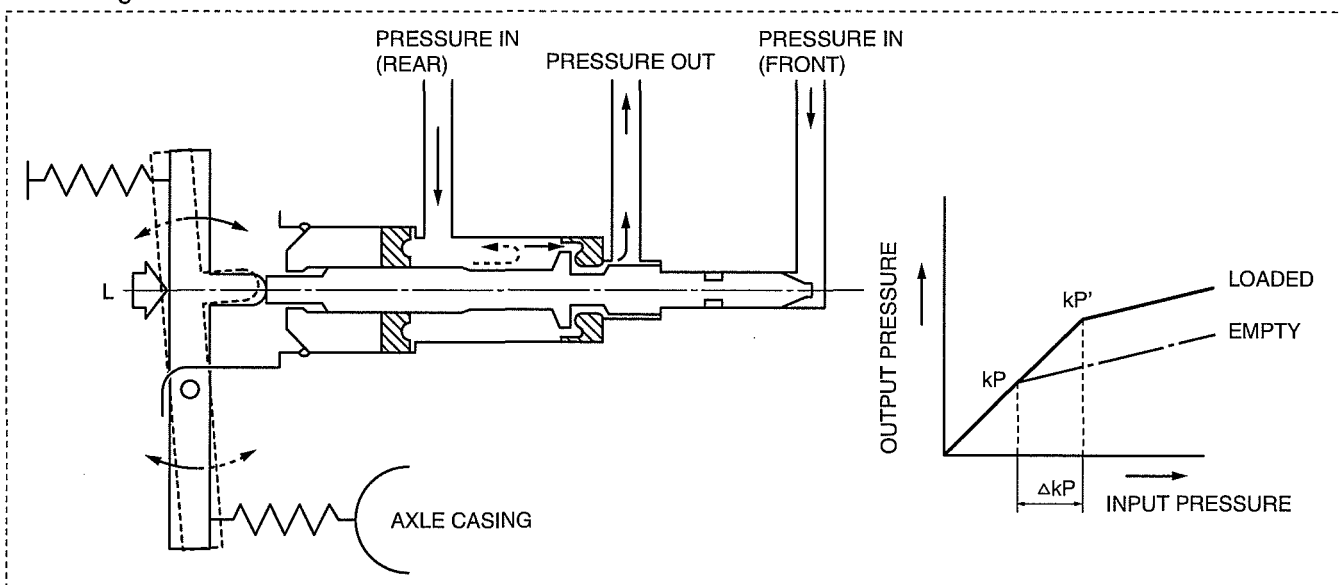
- When the input pressure passes the break point, the plunger is moved back against the cylinder base. The output port is opened, and hydraulic pressure generated at the rear section of the master cylinder is applied to the rear wheel cylinders. As the output pressure continues to increase, the plunger is again moved toward the left against the ring valve, thus separating the input and output side again. This action is repeated to control the output pressure.



DBR411ZTB011

Braking When Vehicle Loaded

- The vehicle height (from the ground) decreases when cargo is loaded into the vehicle. This causes the force of the spring connected to the axle casing to decrease and the force of the spring on the LSPV to increase force L , increasing the pressure holding the plunger to the right. The master cylinder hydraulic pressure (input) needed to cause the plunger to move back to the left also becomes equally higher as a result. This causes the break point to be increased and extra braking power is provided in accordance with the weight of cargo loaded.

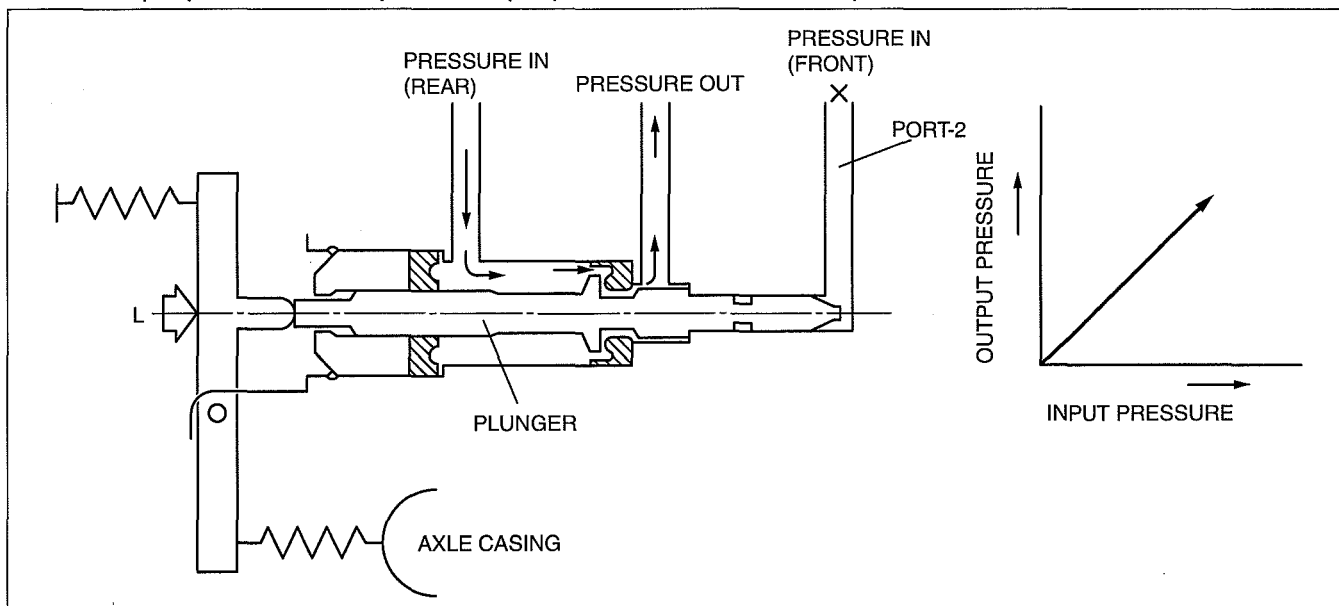


DBR411ZTB012

CONVENTIONAL BRAKE SYSTEM

If Malfunction Of Front Hydraulic System

- If there is a malfunction of the front-brake hydraulic system, the hydraulic pressure from input port 2 becomes zero. The result is that the input pressure presses the plunger to the right, opening the bypass passage. The output pressure thus equals the input pressure, and there is no pressure reduction.



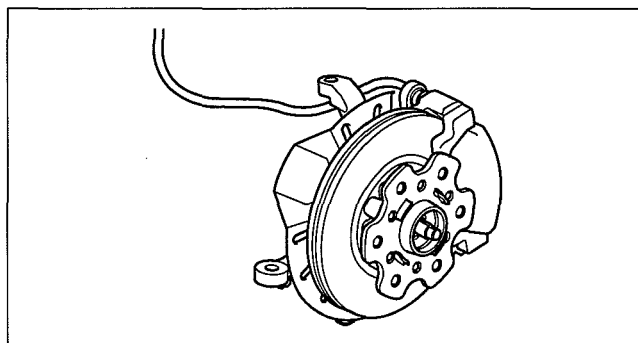
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FRONT BRAKE (DISC) CONSTRUCTION

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4x2 (Except Hi-Rider)

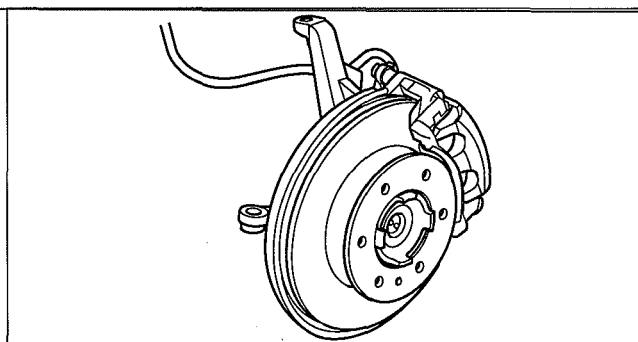
- Large diameter, ventilated disc type front brakes with a **256 mm {10.1 in}** diameter and **24 mm {0.94 in}** thickness have been adopted, improving braking force and fade resistance.



DBR411ZTB004

Hi-Rider, 4x4

- Two-piston type front disc brake caliper has been adopted, improving braking force.
- Large diameter, ventilated disc type front brakes with a **289 mm {11.4 in}** diameter and **28 mm {1.1 in}** thickness have been adopted, improving braking force and fade resistance.



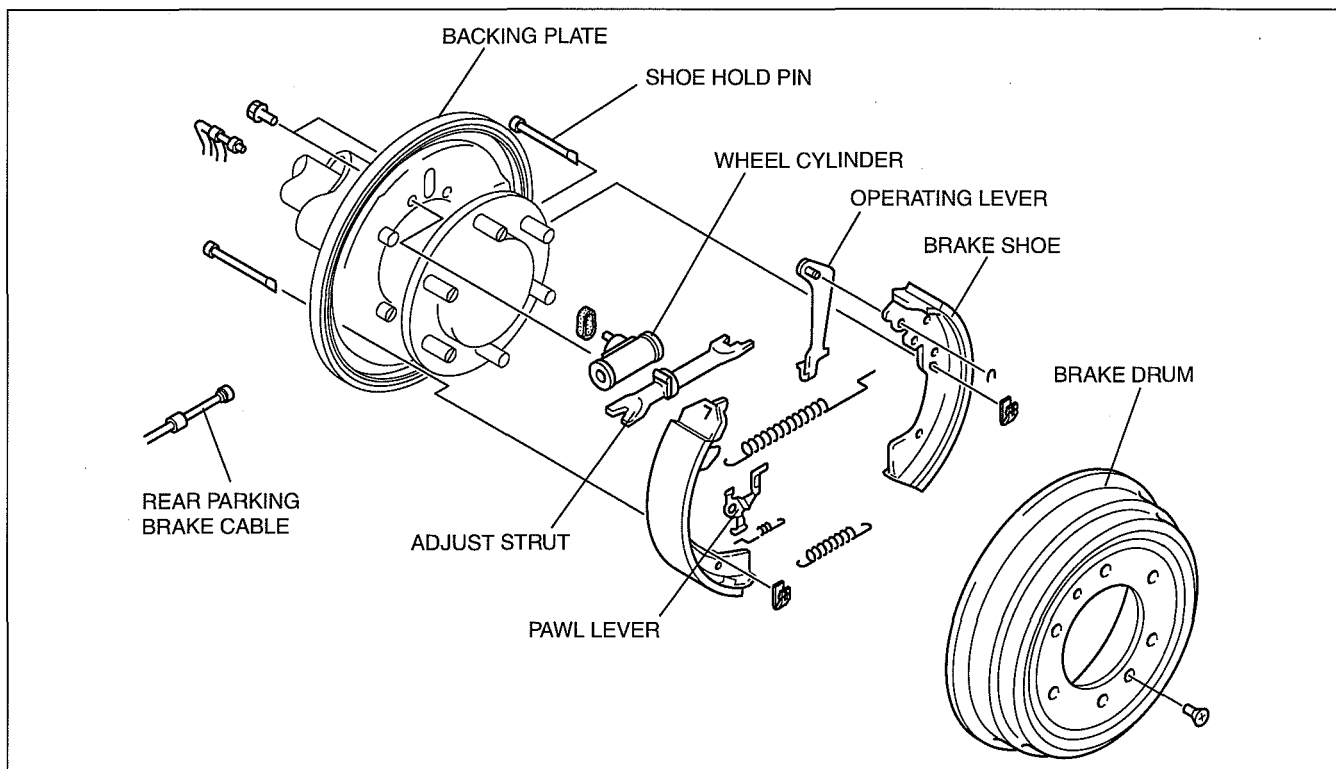
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CONVENTIONAL BRAKE SYSTEM

REAR BRAKE (DRUM) CONSTRUCTION

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- A leading/trailing type rear drum brake, with a **270 mm {10.6 in}** (4x2 (except Hi-Rider)) or **295 mm {11.6 in}** (Hi-Rider, 4x4) drum inner diameter and **23.81 mm {0.937 in}** wheel cylinder inner diameter, has been adopted, improving braking force and fade resistance.
- Braking stability is ensured by use of **55 mm {2.16 in}** width linings.
- If has an incremental automatic adjustment mechanism of shoe clearance to improve serviceability.



DBR411ZTB006

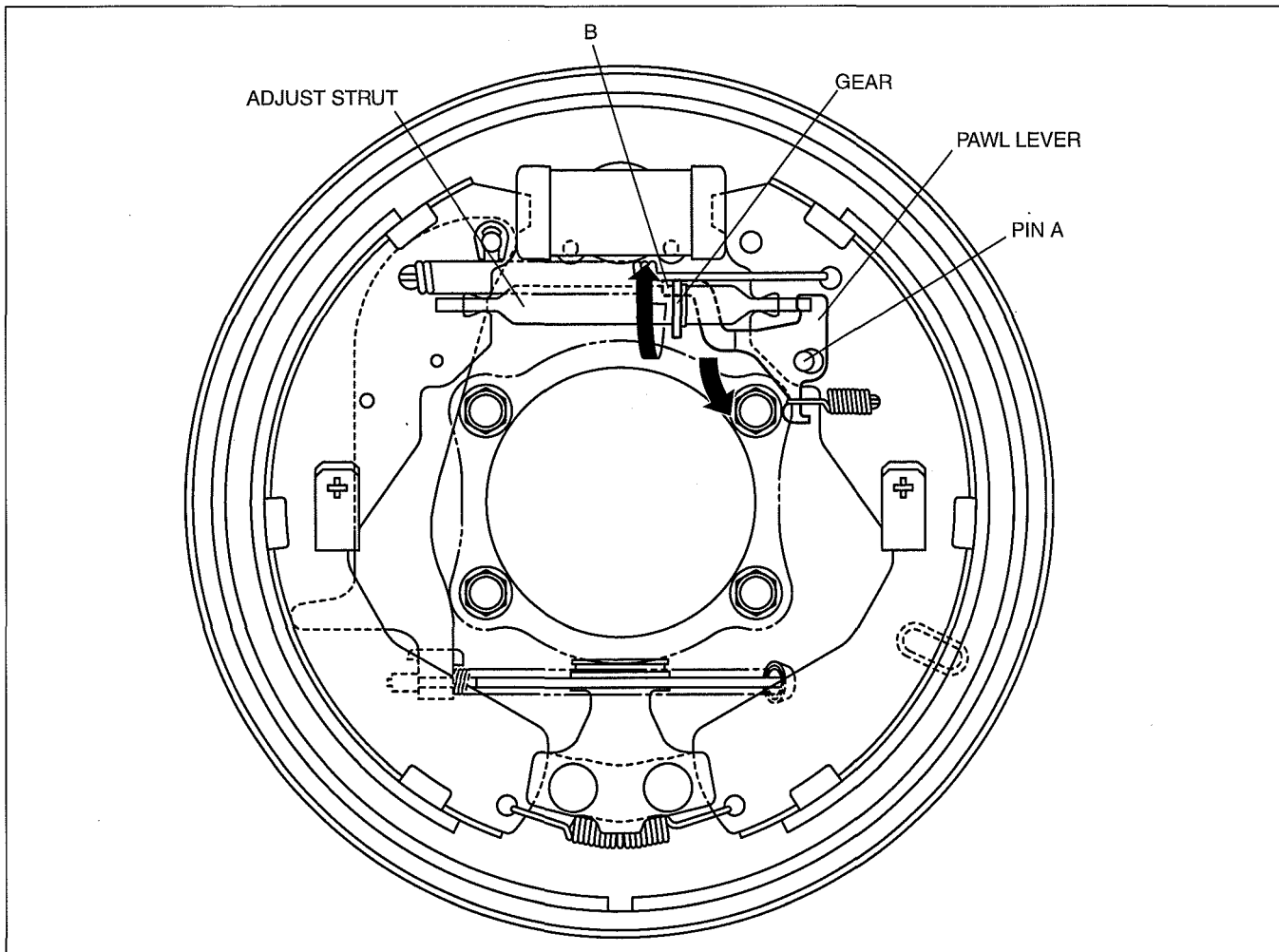
CONVENTIONAL BRAKE SYSTEM

REAR BRAKE (DRUM) OPERATION

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Automatic Adjustment Mechanism

- This brake is equipped with an auto adjust mechanism, which automatically adjusts shoe clearance when the brake pedal is depressed. The auto adjust mechanism is composed of an adjust strut and pawl lever.



DBR411ZTB007

When there is large clearance

- The end of the adjust strut receives the rotation force against the pawl lever caused by the spring which has pin A as its fulcrum. When the brake pedal is depressed, the pawl lever moves in the direction of the arrow with pin A as the fulcrum, because the length of the adjust strut does not change in response to the increase in shoe space.
- The gear rotates because the end B of the pawl lever is engaged with the gear of the adjust strut. Because the gears and the body are engaged by the bolt and nut, the total length of the adjust strut increases when the gear rotates in the direction indicated in the figure. When there is a large clearance, the gear rotates one notch each time the brake pedal is depressed.

When clearance is proper

- When the clearance is proper, clearance is maintained because the shoe does not stroke (the pawl lever does not tilt) enough to rotate the gear one notch, even when the brake pedal is depressed.

PARKING BRAKE SYSTEM

04-12 PARKING BRAKE SYSTEM

PARKING BRAKE SYSTEM OUTLINE. . 04-12-1

PARKING BRAKE SYSTEM
STRUCTURAL VIEW 04-12-1

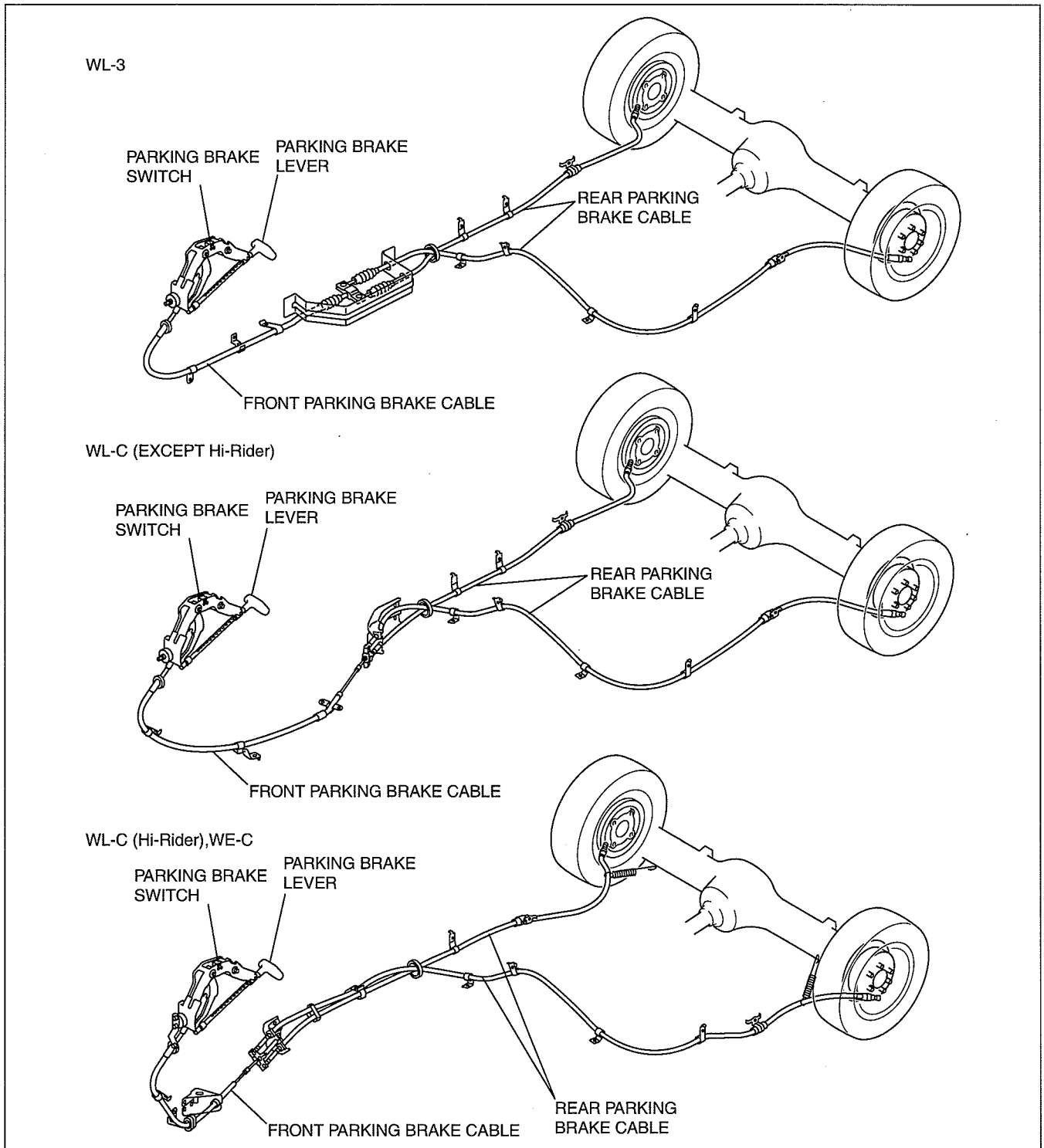
PARKING BRAKE SYSTEM OUTLINE

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- A stick lever type parking brake has been adopted, improving operability.

PARKING BRAKE SYSTEM STRUCTURAL VIEW

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DCF412ZTB001

04-13B ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ABS OUTLINE [4W-ABS]	04-13B-1	ABS CONTROL OPERATION	
ABS STRUCTURAL VIEW [4W-ABS] ..	04-13B-1	[4W-ABS]	04-13B-6
ABS SYSTEM WIRING DIAGRAM		ABS WHEEL-SPEED SENSOR AND	
[4W-ABS]	04-13B-2	ABS SENSOR ROTOR FUNCTION	
ABS HU/CM CONSTRUCTION		[4W-ABS]	04-13B-6
[4W-ABS]	04-13B-2	ABS WHEEL-SPEED SENSOR AND	
ABS HU PART FUNCTION [4W-ABS] ..	04-13B-3	ABS SENSOR ROTOR	
ABS HU PART CONSTRUCTION/		CONSTRUCTION/OPERATION	
OPERATION [4W-ABS]	04-13B-3	[4W-ABS]	04-13B-6
ABS CM PART FUNCTION [4W-ABS] ..	04-13B-5	G SENSOR FUNCTION [4W-ABS]	04-13B-7
ABS CONTROL OUTLINE [4W-ABS] ..	04-13B-5	G SENSOR CONSTRUCTION/	
		OPERATION [4W-ABS]	04-13B-7

ABS OUTLINE [4W-ABS]

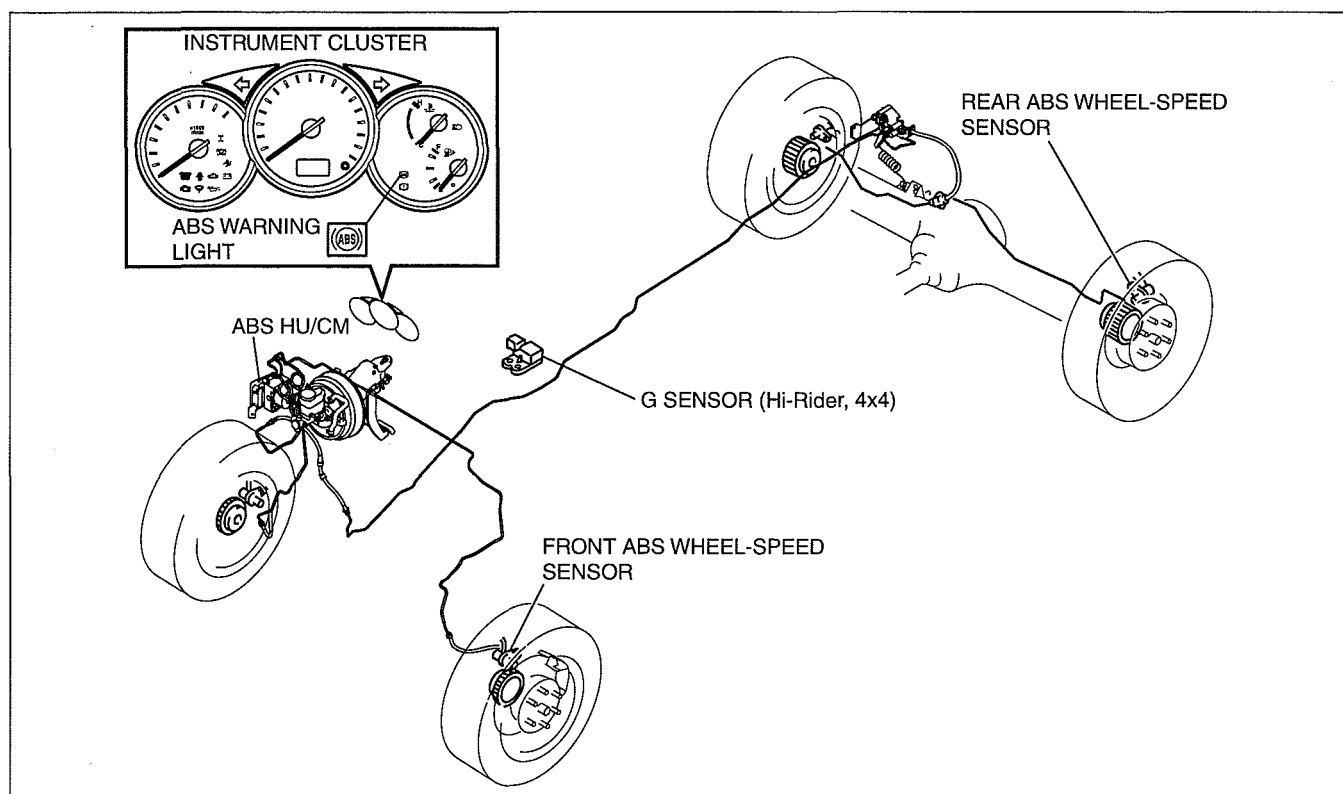
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- The ABS HU/CM, integrating both the hydraulic unit (HU) and control module (CM), has been adopted, resulting in size and weight reduction.
- Select-low controlled-4-wheel anti-lock brake system with 4-sensor and 3-channel has been adopted, and has the following features.
 - The integrated ABS Hydraulic Unit/Control Module (HU/CM) system is compact, lightweight and highly reliable.
 - Combined use of the current diagnostic tool and the ABS HU/CM allows the following functions to improve serviceability:
 - 4-digit DTC display
 - PID/DATA monitor function
 - Active command modes function

04

ABS STRUCTURAL VIEW [4W-ABS]

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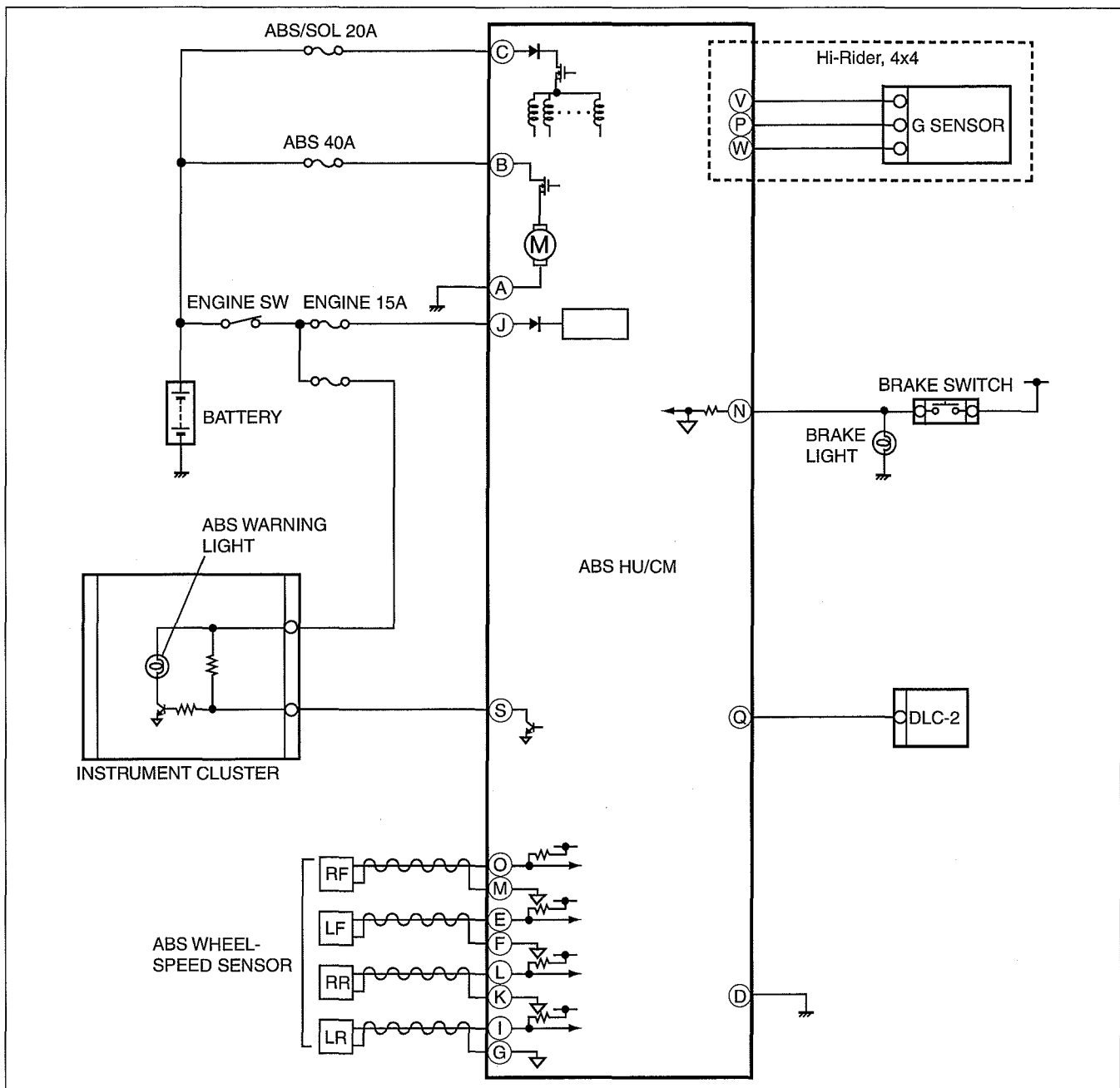


DCF413BTB001

ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ABS SYSTEM WIRING DIAGRAM [4W-ABS]

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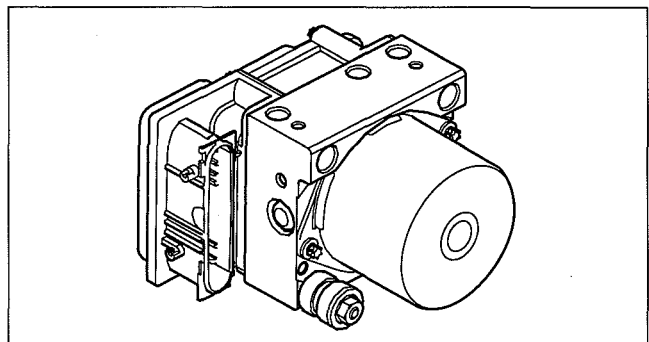


DCF413BW001

ABS HU/CM CONSTRUCTION [4W-ABS]

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- A high reliability, reduced size and weight ABS HU/CM, integrating both the ABS HU and ABS CM, has been adopted.



DBR413ZTB002

ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ABS HU PART FUNCTION [4W-ABS]

dcf041343750t09

- The ABS HU adjusts the fluid pressure to the caliper pistons by controlling (on/off) each solenoid valve and pump motor according to signals from the ABS CM.

ABS HU PART CONSTRUCTION/OPERATION [4W-ABS]

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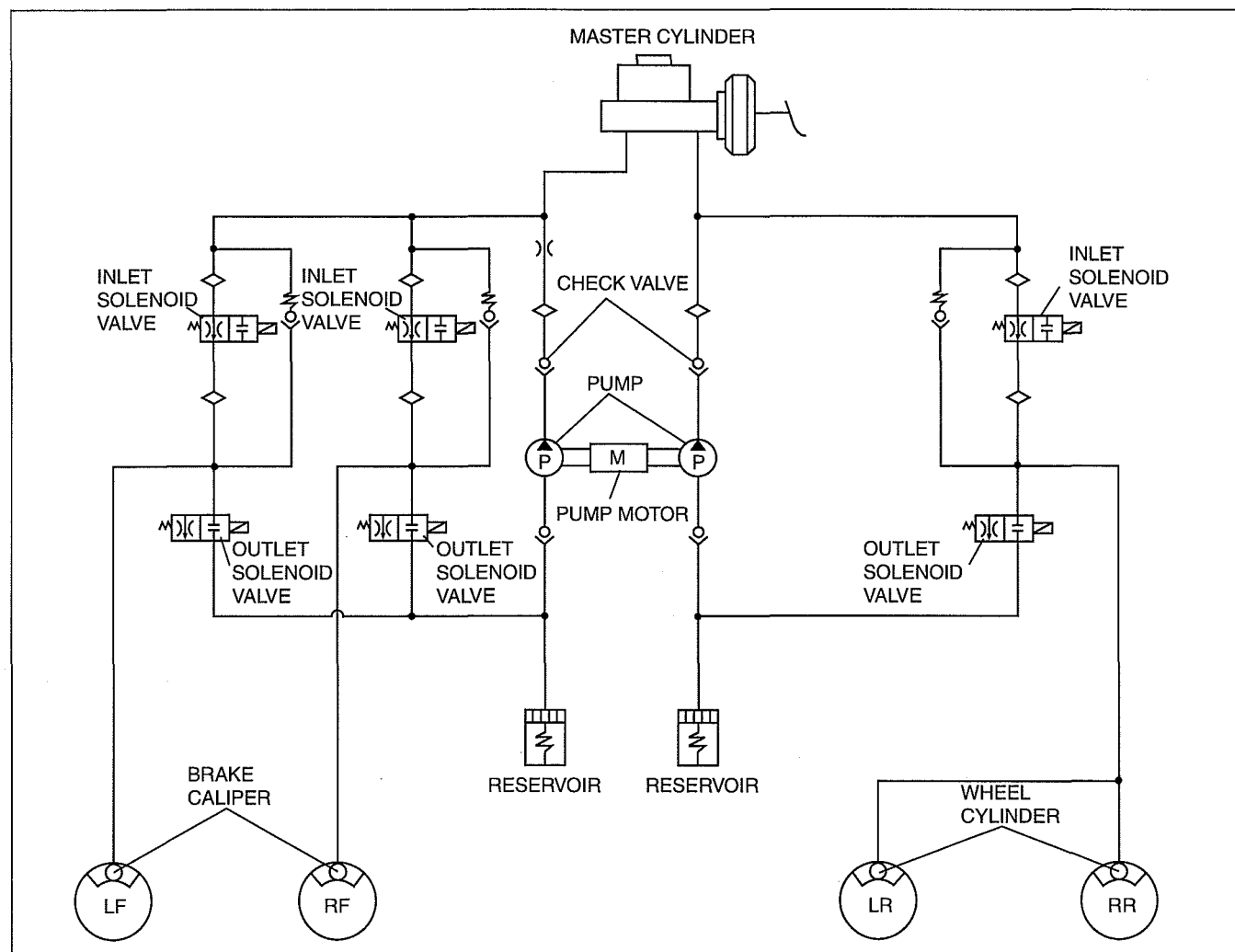
Construction

- The integrated ABS HU/CM system controls the ABS. The ABS are an independent front wheel control, rear axle control, 4-sensor, 3-channel system.

Function of main component Parts

Part name	Function
Inlet solenoid valve, outlet solenoid valve	- The solenoid valves control the fluid pressure at individual brake calipers (wheel cylinders) in accordance with the signal received from the ABS CM. - There are six solenoid valves in total and two solenoid valves in each wheel (one for retaining pressure and one for reducing pressure) are used. There are four valves for the front wheel, and the left and right wheels are controlled separately. There are two valves used to control the rear wheels simultaneously.
Reservoir	The reservoirs temporarily store the hydraulic pressure which flows from the brake caliper (wheel cylinder) through the outlet solenoid valve during pressure reduction.
Pump	The pump returns brake fluid from the reservoir to the master cylinder.
Pump motor	Operates the pump according to ABS CM signals.

Hydraulic circuit diagram



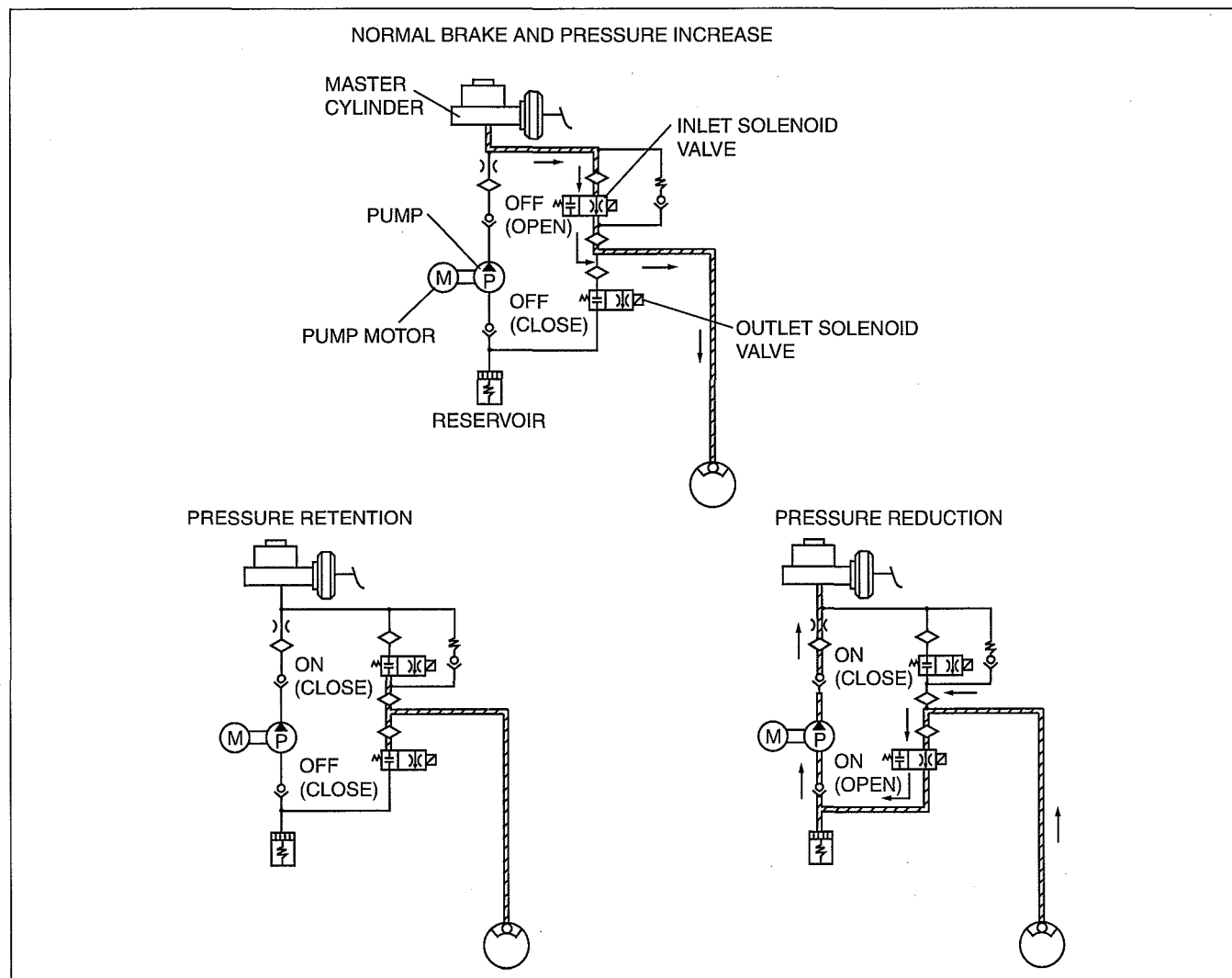
DBR413ZTB003

ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

Operating

- The ABS HU/CM detects wheel-speed and driving condition based on the ABS wheel-speed sensors, and outputs on/off electronic signals to the inlet solenoid valves and outlet solenoid valves to control the ABS control.
- The ABS HU/CM controls ABS by calculating wheel slip from ABS wheel-speed sensor signals, and adjusting the brake pressure.

	Solenoid valve (Inlet)	Solenoid valve (Outlet)	Pump motor, pump
Normal brake	Off (open)	Off (close)	Stopped
Pressure reduction	On (close)	On (open)	Operating
Pressure retention	On (close)	Off (close)	Stopped
Pressure increase	Off (open)	Off (close)	Stopped



DBR413ZTB004

ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ABS CM PART FUNCTION [4W-ABS]

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- The ABS CM detects the vehicle wheel speeds based on the signals from the four ABS wheel-speed sensors. The CM calculates the rotation condition of each wheel from the relation between the detected vehicle wheel speed and the estimated (based on the detected speed) vehicle speed from there on. It then accordingly controls brake fluid pressure to each wheel to prevent lock-up.

Function Table

Function name	Contents
ABS control function	Controls brake fluid pressure when braking to maintain directional stability, ensure steerability and reduce stopping distance.
On-board diagnostic system	- Main components of the ABS control system have a self-diagnosis function. In case a malfunction occurs, warning lights illuminate to alert the driver, and at the same time a DTC is stored in the ABS HU/CM. - When a malfunction is determined as a result of on-board diagnosis, system control is suspended or limited to prevent any dangerous situation while driving.

ABS CONTROL OUTLINE [4W-ABS]

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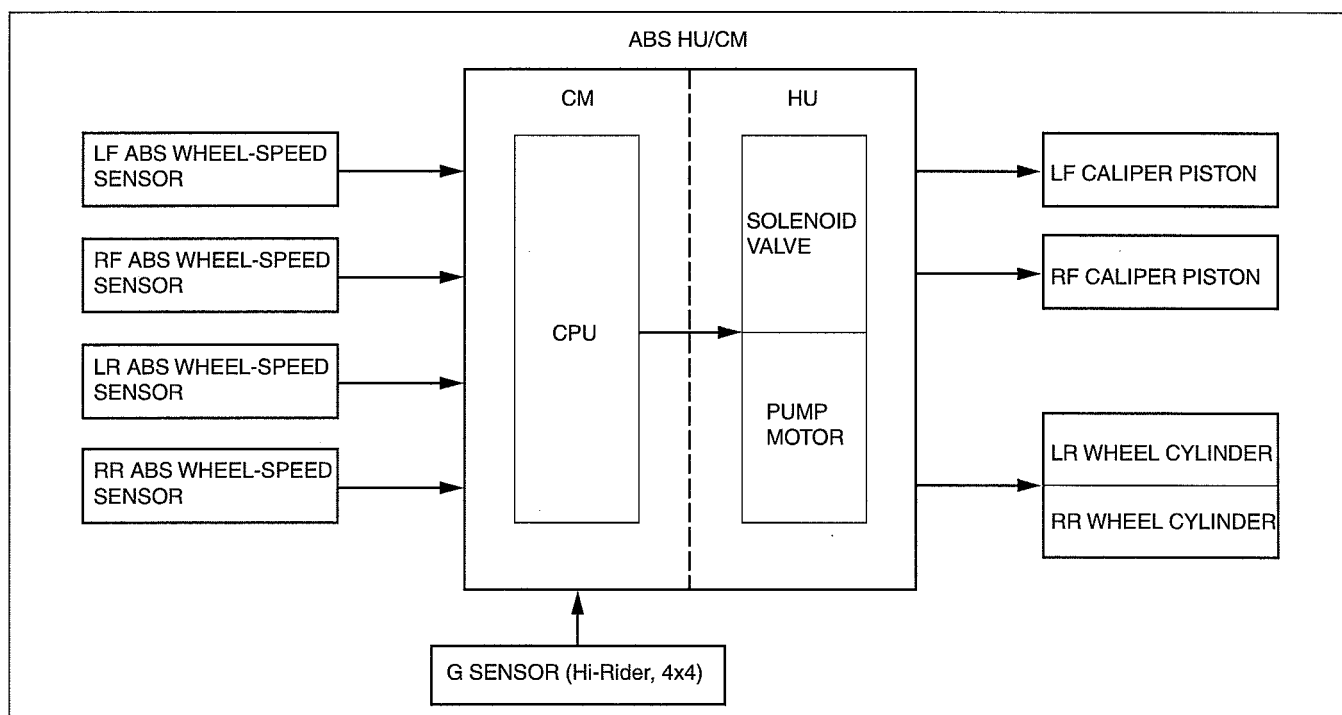
- ABS control occurs when wheel slip is determined by the ABS CM (based on the four ABS wheel-speed sensors). Then, the ABS HU inlet and outlet solenoid valves are operated and brake fluid pressure is controlled accordingly to prevent wheel lock-up.
- Use of ABS control during emergency braking or on slippery road surfaces allows directional stability to be maintained, steerability ensured and stopping distance to be reduced.
- The ABS control system has independent front wheel control and unified control (select low) for the rear wheels.

Note

- Select low control: A control system in which the left and right vehicle wheel speeds are compared and brake fluid pressure is controlled according to the wheel most likely to lock-up.

04

Block Diagram



DCF413ZTB012

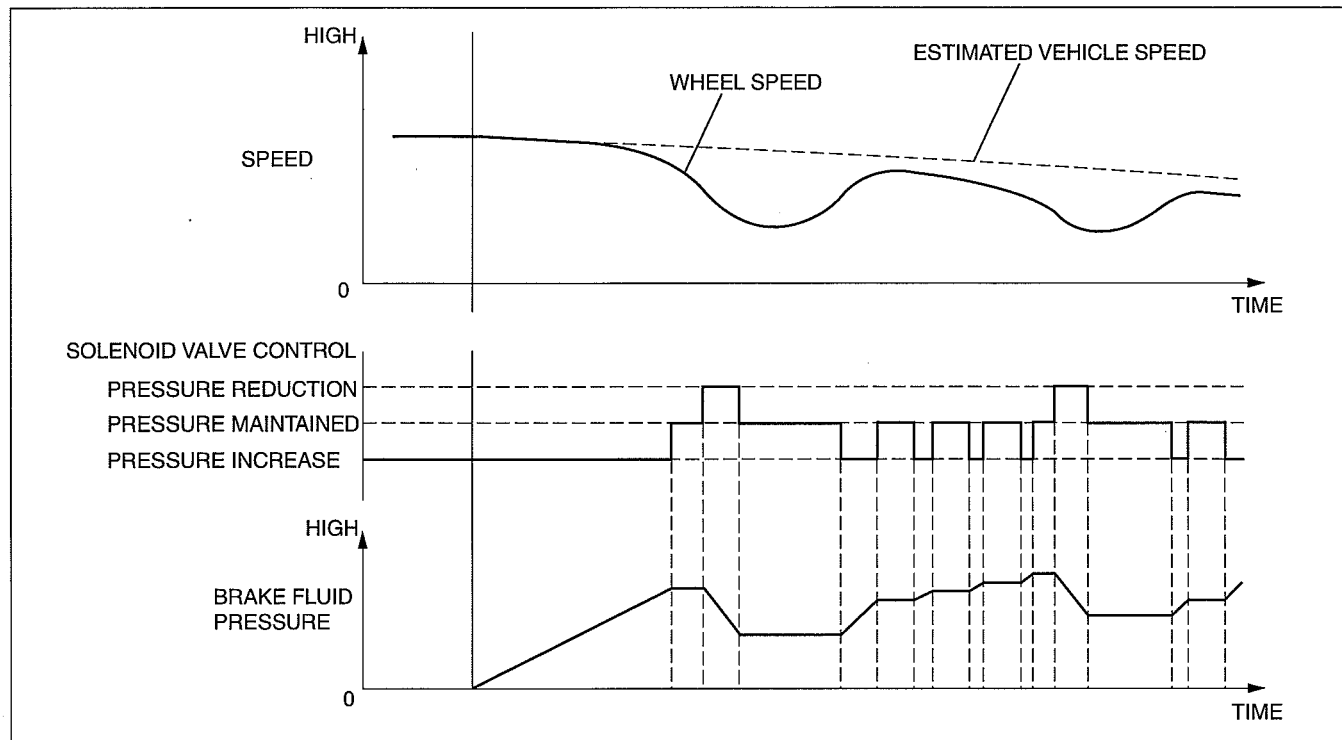
ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

ABS CONTROL OPERATION [4W-ABS]

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- When the ABS CM determines wheel slip conditions based on the signals from the ABS wheel-speed sensors during braking, the ABS CM operates the ABS HU inlet and outlet solenoid valves, reducing and maintaining brake fluid pressure in accordance with the wheel slip factors. Then, when the wheel slip condition has passed, brake fluid pressure is increased and maintained, ensuring braking with a constantly stable brake force.

Operating Condition Transition Diagram



DBR413ZTB012

ABS WHEEL-SPEED SENSOR AND ABS SENSOR ROTOR FUNCTION [4W-ABS]

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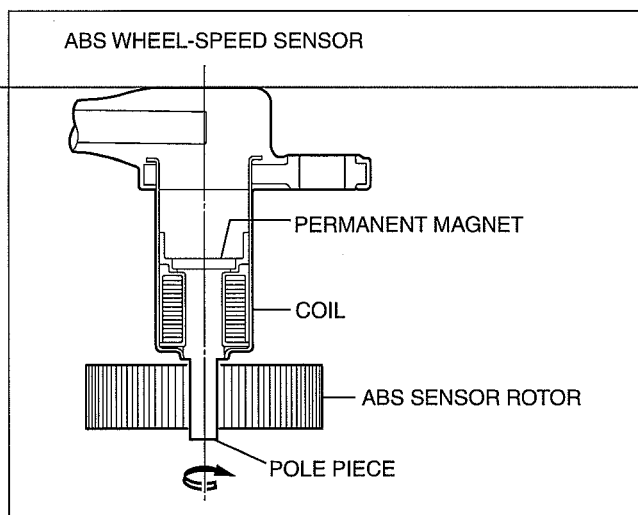
- The ABS wheel-speed sensor, which has a magnetic pick-up, transmits the rotation condition of each wheel to the ABS HU/CM.

ABS WHEEL-SPEED SENSOR AND ABS SENSOR ROTOR CONSTRUCTION/OPERATION [4W-ABS]

dcf041343720t04

Construction

- The front ABS wheel-speed sensor is installed on the steering knuckle and the front ABS sensor rotor is integrated with the front wheel hub (4x2 except Hi-Rider) or drive shaft (Hi-Rider, 4x4).
- The rear ABS wheel-speed sensor is installed on the hub spindle and the rear ABS sensor rotor is integrated with the rear axle shaft.

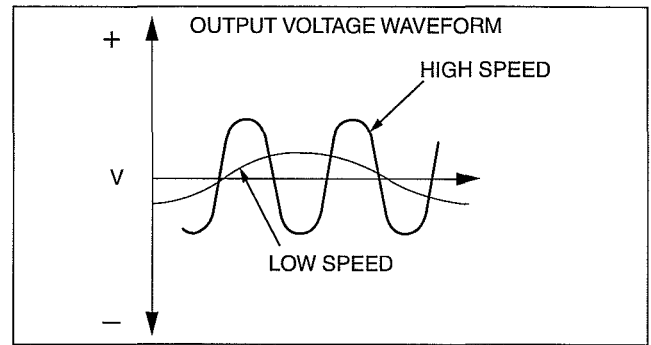


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ANTILOCK BRAKE SYSTEM [4-WHEEL ANTILOCK BRAKE SYSTEM (4W-ABS)]

Operation

- As the ABS sensor rotor rotates, magnetic flux formed from the permanent magnet varies and alternating current is formed with an electromagnetic conductor. Using this alternating current, rotation speed is expressed as a varying proportional cycle and from detection of this cycle the CM part of the ABS HU/CM can then detect the wheel rotation speed. While the structures of the front and rear ABS wheel-speed sensor differ, the operation is the same.



DBR413ZTB011

G SENSOR FUNCTION [4W-ABS]

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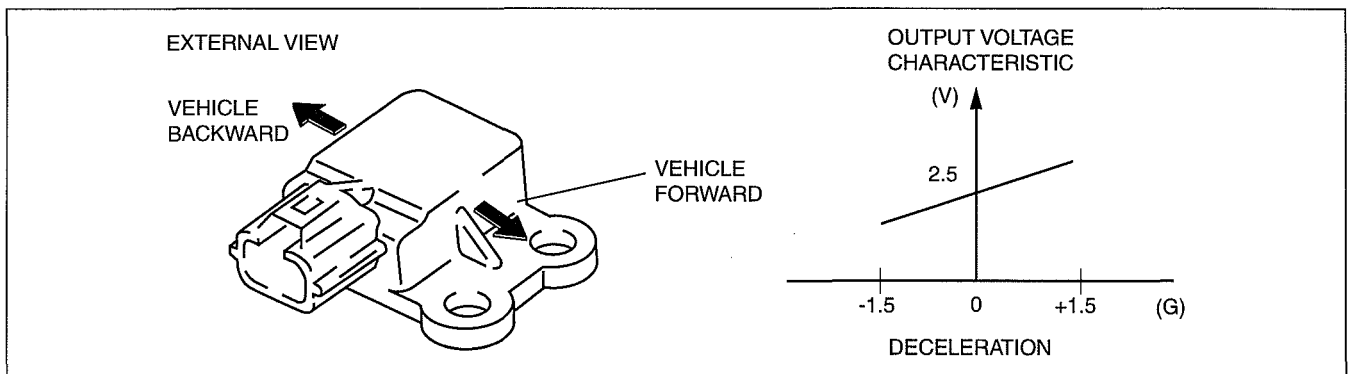
Note

- Because the Hi-Rider vehicle is based on the 4x4 vehicle, the 4x4 vehicle ABS system has been adopted. Therefore, the G-sensor, which is generally installed to 4x4 vehicles only, is installed to the Hi-Rider.
 - Because the 4x4 vehicle ABS system has been adopted on the Hi-Rider, the ABS is controlled in the same way as the 4x4 vehicle.
- In the Hi-Rider and 4x4 vehicle, the G sensor detects and calculates the vehicle G force (acceleration and deceleration G force), and outputs the change in voltage to the ABS HU/CM.

G SENSOR CONSTRUCTION/OPERATION [4W-ABS]

dcf041343770t02

- The G sensor is installed under the RH front seat and detects vehicle movement during a brake lockup event that is transferred to other wheels through the powertrain, and transmits the signal to the ABS HU/CM.
- The drive train of 4x4 vehicles delivers driving force to all four wheels, and due to this interlocking, the speed of all wheels during braking is the same. When driving on road surfaces with especially low μ (friction coefficient), it is difficult to estimate the vehicle speed based on the wheel speed, and ABS braking becomes unreliable. (For 4x2 vehicles, the front and rear wheels are independent, so it is possible to accurately estimate the vehicle speed by measuring the difference between the rotation speeds of the front and rear tires.) A G sensor is installed to overcome this situation. When braking, the change in G-force is detected, and the road surface μ is judged (low μ , high μ), enabling a correction of the vehicle speed estimation.
- The output characteristics are based on 2.5 V with the vehicle in a stopped or driven with constant speed, and the subsequent fluctuation is depending on the situation whereby the G force occurs.



DBR413ZTB006

(1)

(2)

(3)